

Tengteng Niu

+86 13366655407 | ntt21@mails.tsinghua.edu.cn | TNTmohan.github.io | linkedin.com/in/tengteng-niu

Education

- Tsinghua University**, Ph.D. in Electrical Engineering (expected June 2026) Sept 2021 – June 2026
- **Research interests** : AI-assisted materials discovery, insulation for wide-bandgap power electronic devices.
 - **Award**: Outstanding Teaching Assistant Award, 2023; Oriental Cable Talent Scholarship, 2023
- Tsinghua University**, B.E. in Electrical Engineering Sept 2017 – June 2021
- **Courses**: Digital/Analog Circuit Analysis, Fundamentals of Power Electronics, Principles of Automatic Control, Microcomputer Principles, Signal System Analysis, Power System, High Voltage Engineering
 - **Award**: Comprehensive Excellence Award, 2018; Academic Excellence Award, 2020
- Tsinghua University**, Minor in Data Science and Technology Sept 2018 – June 2021
- **Courses**: Data Structures, Database Technology, Big Data Systems, Machine Learning, Deep Learning

Technical skills

- **Programming Languages**: Python, Java, C/C++ , Matlab, JavaScript, HTML, CSS
- **Software/Frameworks**: PyTorch, scikit-learn, RDKit, COMSOL Multiphysics Simulation

AI Projects

Performance Prediction Model for Crosslinked Polymers Using Graph Convolutional Neural Networks

- Represented polymer crosslinked structures with weighted undirected graphs to capture crosslinking information absent in traditional molecular representations.
- Applied transfer learning with large-scale pre-training and small-sample fine-tuning, reducing training data requirements by $5\times$ under the same RMSE.

High-Performance Molecular Inverse Design Framework Based on Actor-Critic Reinforcement Learning

- Designed GRU- and Transformer-based policy networks to construct molecular structure generators.
- Developed a reward function integrating molecular synthesizability evaluation; optimized molecular generation via policy gradient training.

Reinforcement Learning-Based Optimal Power Flow Algorithm for Power Systems

- Modeled OPF using the A3C algorithm to minimize total network losses under bus voltage and generator output constraints.
- Achieved $>12\times$ faster computation than conventional methods on IEEE 9-bus system (672 test cases) with only +0.18 MW average loss.

Projects

High-Voltage High-Frequency Wide-Bandgap Power Electronics Packaging Design

- Investigated insulation failure mechanisms under HV (>10 kV) and HF (>10 kHz) conditions.
- Built an HF partial discharge acquisition platform and developed adaptive wavelet denoising algorithms.

1500V High-Voltage Energy Storage Battery System and Key Power Conversion Technologies

- Investigated insulation failure and arc fire risks in high voltage energy storage systems; developed a battery system electric field simulation model that spans 4 orders of magnitude (10^{-3} – 10^1 m).
- Clarified the impact of common-mode interference from high-frequency power electronic switches on battery pack insulation failure.

Fault Characteristics and Evolution of Key Components in Energy Storage Battery Systems

- Developed coupled thermo-electro-mechanical multiphysics models for key components (battery system, PCS, transformer, BMS).
- Extracted critical fault parameters, identified multilevel fault features and propagation paths, and achieved fault decoupling across system levels.

Publications

Journal Articles

- [1] **T. Niu**, Q. Yi, Z. Zhang, and Y. Yang, "Interface Engineering for Synergistic Achievement of Enhanced Thermal Conductivity and Suppressed Dielectric Loss of Epoxy/Boron Nitride Composite," *High Voltage*, 2025. (Accepted, Cover Article)
- [2] **T. Niu**, Z. Zhang, and Y. Yang, "Frequency-Dependent Breakdown Strength of Epoxy Composites: Role of High Thermal Conductivity and Low Dielectric Loss," *IEEE Transactions on Dielectrics and Electrical Insulation*, 2025. (Accepted, Preprint)
- [3] **T. Niu**, R. Huang, Z. Qu, D. Hui, and Y. Yang, "1500 V Li-ion Battery Pack Electric Field Simulation and Insulation Risk Analysis," *Proceedings of the CSEE*, 44, 377–385, 2024. (in Chinese)
- [4] **T. Niu**, Y. Du, Y. Fang, and Y. Deng, "Effect of Voltage Ramp Rate on Breakdown Characteristics of Epoxy under High-Frequency Bipolar Square Wave," *High Voltage Engineering*, 2025. (Accepted, in Chinese)
- [5] T. Yao, K. Chen, **T. Niu**, and Y. Yang, "Effects of Frequency and Thermal Conductivity on Dielectric Breakdown Characteristics of Epoxy/Cellulose/BN Composites Fabricated by Ice-Templated Method," *Composites Science and Technology*, 213, 108945, 2021.
- [6] T. Yao, C. Zhang, K. Chen, **T. Niu**, J. Wang, and Y. Yang, "Hydroxyl-Group Decreased Dielectric Loss Coupled with 3D-BN Network Enhanced High Thermal Conductivity Epoxy Composite for High Voltage–High Frequency Conditions," *Composites Science and Technology*, 234, 109934, 2023.
- [7] T. Yao, C. Zhang, Z. Zhang, **T. Niu**, Q. Yi, and Y. Yang, "Hydrogen Bonding and $\pi - \pi$ Interaction Enhanced High Thermal Interface Design with Low Dielectric Loss of BN/PI/Epoxy for High Voltage–High Frequency Insulating Applications," *Advanced Composites and Hybrid Materials*, 7, 30, 2024.

Patents

- [1] Y. Yang, T. Yao, K. Chen, **T. Niu**, and J. Wang, "Epoxy Resin-Based Composites and Their Preparation Method," China Patent No. 202110328465X, Invention Patent.
- [2] Y. Yang, T. Yao, K. Chen, **T. Niu**, and J. Wang, "Composite Materials and Their Preparation Method," China Patent No. 202110330913X, Invention Patent.
- [3] Y. Yang, **T. Niu**, T. Yao, and J. Wang, "High Thermal Conductivity and High Insulation Epoxy Resin Composites and Their Preparation Method," China Patent No. 202211540332X, Invention Patent.